Yash Raj

+91-8340272911 | yashrajalways4u@gmail.com | https://github.com/23f2004827| linkedin.com/in/yash-raj-656a59287/

EDUCATION

Indian Institute of Technology Madras *Chennai, India* **Bachelor of Engineering** in Data Science and Applications 2023 – 2027

- Among the top 10% of the batch
- Relevant coursework in [Relevant Subjects: Machine Learning, Deep Learning and Generative AI, DSA(Python), JAVA, Tools in Data Science, Business Data Management,]

SKILLS

Language: Python, SQL, Java, JavaScript

AI/ML: PyTorch, Scikit-learn, Hugging Face Transformers, XGBoost, LightGBM, CatBoost, LoRA Fine-tuning

Generative AI: LLMs, RAG, Transformer Models, Prompt Engineering, Embeddings, Semantic Similarity

Backend: Flask, FastAPI, Django, REST APIs

Databases: SQLite, Redis

Tools: Git, GitHub, GCP, Weights & Biases, Kaggle, Excel, dbt,Power BI, Tableau

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, REST APIs

CERTIFICATIONS

- WebDev, DevOps and Blockchain Cohort– Feb'2025
  (https://generateinvoice.teachx.in/generatecertificate/certificate/harkirat_db/100023/51270/14)
- AI & Entrepreneurship: Building a Successful AI-First Product Company– Aug'24
  (https://drive.google.com/file/d/1Pte3hLb5c66e_ABiXz6Y5AUtkT5p27Kv/view?usp=sharing)
- Foundation Level in Data Science and Application– Dec'24
  (https://drive.google.com/file/d/1ZXaPqBD-ZmOKg8ITwwBJ_6VXe8Cj4x9k/view?usp=sharing)
- Diploma in Programming – Dec'25
  (https://drive.google.com/file/d/1KduCycwvWFEyLk_PeeG9NuZYoQ3cyUhO/view?usp=sharing)

PROJECTS

LLM-Based MCQ Answer Ranking System | Python, Hugging Face Transformers, PyTorch Scikit-learn, Weights & Biases, Kaggle

• Built an end-to-end LLM pipeline to predict and rank the top three answers for challenging multiple-choice questions using transformer-based language models.
• Implemented text preprocessing, semantic similarity, transformer embeddings, LoRA fine-tuning, and retrieval-augmented techniques to improve answer ranking performance.
• Tracked experiments with Weights & Biases and evaluated multiple model configurations using MAP@3 to benchmark performance.
Developed reproducible training, inference, and evaluation workflows with GitHub version control and Kaggle-based benchmarking.

Placement Portal Application | Flask, REST APIs, Redis, Celery, SQLite

- Developed a full-stack placement management system with **role-based access**, **REST APIs**, **Redis caching, and asynchronous Celery jobs**.
- Implemented database-driven workflows for Admin, Company, and Student users.

Heavy Equipment Price Prediction | Python, Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, Kaggle

- Developed an end-to-end machine learning pipeline to predict heavy equipment selling prices using structured operational, transactional, and technical features from a large-scale Kaggle dataset.
- Performed comprehensive **EDA**, feature engineering, missing-value handling, categorical encoding, feature scaling, and reusable preprocessing pipelines to improve model performance.
- Trained and compared multiple regression models including **XGBoost, LightGBM, CatBoost, and Random Forest**, followed by hyperparameter tuning and model evaluation using **RMSLE**.
- Analyzed feature importance, prediction errors, and model performance to improve leaderboard results.

ACHIEVEMENTS

- Qualified GATE Computer Science Engineering (CS) 2026
- Qualified GATE Data Science and Artificial Intelligence (DA) 2026
- Achieved Grade A in Machine Learning Techniques